Specific Aims

Craniosynostosis is the early fusion of the cranial sutures and affects 1 in 2,100 live births [1]. Children with this condition typically undergo surgical treatment to remove brain growth constraints and correct for the malformations produced by a volume overgrowth parallel to the fused sutures that compensates for the locally constrained growth [2]. However, there is a large variability of surgical approaches and results [3–6], and institutions have reported up to 23% [7] of patients with unsatisfactory outcomes [4,8,9] who may require additional surgeries to prevent or treat potential sequalae associated with development of intracranial hypertension [10,11]. These suboptimal treatments are common for three main reasons: (1) local volume anomalies and their progression have not been characterized to estimate how much extra local volume patients need during treatment; (2) our knowledge about how treatment modifies cranial growth is limited and post-surgical development predictions are not possible; and (3) there are no objective and personalized methods to evaluate long-term outcomes, so treatment selection remains subjective. **We aim to characterize the processes of abnormal local volume development in patients with craniosynostosis before and after treatment, create methods to evaluate, predict and compare post-surgical outcomes, and identify relapsing patients using objective data**.

The traditional evaluation of craniosynostosis and its treatment are highly dependent on subjective clinical expertise [2,12,13]. Our team and others have created objective methods to quantify cranial shape anomalies, identify craniosynostosis and plan its treatment using CT images [13–17], which we have extended to 3D photogrammetry [18–22]. However, existing methods cannot characterize the abnormal patterns of local volume development in craniosynostosis, are age-agnostic, and do not account for sex, which is an essential modulator of development. Moreover, although we recently created the first fully data-driven cranial bone shape model predictive of normal growth using cross-sectional CT images [23], there are no quantitative methods to characterize local volume growth before or after treatment of craniosynostosis. This limited quantitative knowledge of the pre- and post-surgical local growth patterns has hindered the clinical translation of existing methods to plan, quantify, compare and predict surgical outcomes. Hence, treatment selection remains subjective and the identification of relapses relies on the clinical interpretation of variable symptoms of pressure with low predictive power [24,25].

We have shown that CT and 3D photogrammetry provide the same quantification of head anatomy [18], and we have incorporated pre- and post-surgical 3D photogram acquisition in the standard clinical protocol of patients with craniosynostosis. In this project, we will leverage our retrospective image dataset to characterize local head volume development in patients with craniosynostosis before treatment, quantify how different treatments modify their growth patterns, and explore the feasibility of identifying relapses using image data. Our specific aims are:

Specific Aim 1: **To characterize local head volume distributions in craniosynostosis**. We will build an age- and sex-specific normative statistical model of the head surface anatomical variability using retrospective CT images of 1,480 subjects (age 0-10 years) without cranial pathology. We will design a novel deep geometric neural network to segment and standardize the representation of the head surface from 3D photogrammetry, which will produce the same representations than our existing CT image processing methods [23]. We will use our normative model to quantify abnormal local volume distributions in the 721 pre-surgical images (363 CT images, 358 3D photograms) of 468 patients with craniosynostosis. Then, we will design and solve a structural equation model of the relationships between the volume loss in head areas with constrained growth and the compensatory overdevelopment in other areas. Finally, we will study their progression with age and the role of sex, which we will evaluate using the 371 pre-surgical longitudinal images of 161 patients with craniosynostosis.

Specific Aim 2: **To predict and evaluate local head volume growth after treatment of craniosynostosis**. We will design and train a new deep phenotyping neural network using images of patients with craniosynostosis before treatment and healthy subjects (2,201 images) to predict head growth. The network will create age-independent patient representations and it will learn to combine them with age to predict patient-specific head anatomies. We will predict post-surgical development for 238 patients with craniosynostosis (average of 3.92 longitudinal images/patient). We will compare our predictions with the real image observations to model the post-surgical progression of abnormal local volume and growth patterns, and we will use it to quantify and compare outcomes between endoscopic and cranial reconstruction surgeries. Finally, we will use our longitudinal post-surgical images to train a multi-task neural network to evaluate and predict post-surgical growth, and we will use it to evaluate the feasibility of automatic identification of relapsing patients who needed additional treatments.

At the end of this project, we will have built the first quantitative reference of local head volume development for patients with craniosynostosis based on the quantified correlations between underdeveloped head regions and local compensatory volume overgrowth, which is key to improve treatment planning. We will have also created methods and tools to evaluate, compare and predict post-surgical growth after different treatments, and to identify potential relapses. This project is an essential step towards the optimization of treatment outcomes.

Research Strategy

1. Significance

Cranial volume distributions in craniosynostosis. Craniosynostosis affects 1 in 2,100 live births [1] and its most common presentation are cranial malformations in the first year of life [26]. These malformations are caused by local growth constraints at the areas of suture fusion and a compensatory volume overgrowth parallel to those sutures [27,28]. Hence, despite growth constraints, these patients have shown normal head volumes explained by the compensatory growth mechanisms in the cranium [21,29,30]. While we have developed methods to quantify head shape anomalies [15–18,22,31], the relationships between local volume decrease at regions with constrained growth and volume overgrowth in other areas have not been studied, which is essential to understand abnormal patient development and plan treatments accounting for current and future local brain volume needs. **We will quantify and model for the first time the relationships between local volume underdevelopment caused by suture fusion and the compensatory volume overgrowth parallel to the fused sutures.**

Automatic non-invasive evaluation of cranial pathology. Although 3D photogrammetry is a cheap and safe alternative to computed tomography (CT) imaging, it only depicts the head surface and the bones are not visible. We showed that both CT and 3D photogrammetry provide the same local quantification of the shape anomalies, and that their quantitative evaluation provide the same accuracy evaluating and identifying craniosynostosis [18]. Our clinical team incorporated the acquisition of 3D photogrammetry in the standard protocol for patients with craniosynostosis and we have been routinely acquiring both pre- and post-surgical images of patients during the last few years. However, although our 3D photogrammetry shape analysis methods showed great potential and others have reproduced our approach [32], existing tools to analyze 3D photogrammetry are based on traditional image processing methods that are not specifically designed for this imaging modality [18,33,34], which consists in discrete unstructured meshes instead of voxel-based volumetric images. Hence, subjective manual corrections are often needed and its automatic accuracy is suboptimal. **We propose to create new geometric learning methods designed specifically for non-invasive 3D photogrammetry to quantify local head anomalies**.

Pediatric evaluation from a developmental perspective. As a developmental pathology affecting children, the course of craniosynostosis depends on the stage of development at the onset of early suture fusion [17,35]. Current methods to quantitatively evaluate local cranial anomalies from imaging data are based on the same approaches used to evaluate adult images [18,21,31,32] and hence, they do not account for the fast anatomical changes associated with age and sex during early life. In this proposal, **we will create anatomical head development models as a function of age and sex, and we will quantify the role of these two variables in the severity of the abnormal local volume growth in patients with craniosynostosis before and after treatment.**

Objective evaluation of surgical outcomes: After confirmation of diagnosis using CT images [2,14], patients with craniosynostosis undergo surgical treatment with two main goals [11,36]: (1) removing cranial growth constraints so the cranium can accommodate volume for a normal brain development; and (2) correcting aesthetic malformations. Since traditional imaging modalities are discouraged after treatment to avoid radiation and/or sedation in children [37,38], it has been challenging to objectify the evaluation of longitudinal surgical outcomes, and treatment selection remains largely subjective. Quantitative methods to assess global and local volume improvements after treatment are essential to evaluate if the cranium can accommodate the necessary volume for the brain to resume a normal development. However, although we have successfully used radiation-free 3D photogrammetry to quantify post-surgical shape malformations [18,19] and intra-cranial volume [20,21], our methods could not evaluate local volume growth. This has prevented the objective evaluation of post-surgical development, which is necessary to reduce the subjectivity and variability of treatments and outcomes [3–5]. In this project, we will **quantitatively characterize the pre- and post-surgical volume anomalies and their progression in patients with craniosynostosis, and we will create the first quantitative framework to evaluate, compare and predict long-term outcomes after different surgical treatments**.

Prediction of recovery and early identification of relapses. Since large post-surgical image datasets of patients with craniosynostosis have traditionally not been available, there are no data-driven studies about the post-surgical progression of local head anomalies. Hence, the current identification of unsatisfactory surgical outcomes of patients who may require additional treatments is based on the observation of clinical symptoms of intra-cranial hypertension whose interpretation is subjective and that have a low predictive value [24,25]. Although we have recently created the first data-driven predictive model of pediatric cranial bone development using a large cross-sectional dataset of CT images [23], this model cannot be used to predict the development of a surgically modified cranium. In this project, we will use a unique 3D photogrammetry dataset to **quantitatively evaluate and predict post-surgical head development** in patients with craniosynostosis and we will evaluate the feasibility of automatically **identifying patients who may require additional surgical treatments.**

2. Innovation

Geometric segmentation from craniofacial 3D photogrammetry. Unlike traditional voxel-based image representations, 3D photogrammetry has a variable number of connected vertices with Euclidean coordinates and texture data. Most segmentation and landmark detection methods used to analyze this image modality rely on traditional image processing methods, so they require substantial pre-processing and/or resampling operations [18,19,31,39–43] to convert between representation that are prone to errors, accuracy losses and suffer from cumulated inaccuracies from different steps. Hence, inefficient manual corrections that introduce biases are often necessary. We will develop **a new deep geometric neural network method that will leverage the implicit structure of 3D photogrammetry to automatically segment the head surface around the calvaria**.

Standard anatomical representations across image modalities. Deep learning models trained on multi-modal data can identify modality-specific representations that may cause overfitting and suboptimal results. For that reason, significant data augmentation that introduces appearance variability is often necessary [44–46]. We recently presented a new 2D standard anatomical representation of the calvaria based on spherical coordinates [23] (see Fig. 1) that could reconstruct a cranial anatomy without significant accuracy loss, allowed for data size reductions of 99% compared to CT images, and did not keep modality-specific information. This representation enabled establishing anatomical correspondences between subjects and the development of computationally feasible methods that we used to create the first data-driven predictive model of cranial bone shape during childhood [23]. Based on our previous work, we will train a new **deep geometric network to create standardized representations of the head surface from 3D photogrammetry that will be indistinguishable from the ones obtained from CT images**.

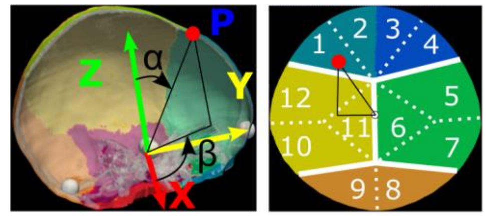

Fig. 1. Standard representation of the cranial surface [23]. A Euclidean point $P(x, y, z)$ in red on the head surface (left) is mapped to a coordinate (α, β) in a spherical map (right). Colors represent cranial bones. Dotted lines separate numbered regions dividing the bones based on the closest suture (shown as white lines).

Separating pathology from anatomical diversity: Principal component analysis (PCA) is commonly used to create anatomical models [14,18,47] and characterize pathology [48–50]. However, it assumes that the population variability is only consequence of a combination of a group of common factors. While this has limited implications when modeling healthy ranges of normally distributed anatomies, its use to characterize pathology is highly limited by its disregard for group differences unrelated to it. In this proposal, we will create a structural equation model that will **quantify the commonly observed patterns of local abnormal volume development in patients with craniosynostosis separately from the unique anatomical traits of every patient**.

Pediatric deep phenotyping independent from the stage of development. Many convolutional neural networks can learn simplified latent representations of complex patient data and generate realistic synthetic observations from the learned distributions [51–56]. However, most only encode the observed phenotypes at the time of imaging and they cannot decouple patient-specific representations from their stage of development. To our knowledge, there are no neural networks that can learn age- and sex- independent patient representations only from cross-sectional data. This is important to evaluate children whose anatomies change rapidly with age and for whom longitudinal studies are not available. For instance. although the convolutional network presented in [57] showed potential predicting Alzheimer's disease progression in adults, it could not infer age-agnostic patient-specific representations. In the current project, we will create a **deep phenotyping framework that will learn patient representations that are independent from age and sex using only cross-sectional images.**

Predictive modeling. While temporal development can be quantified from longitudinal datasets [47,58], these are not generally available for children. For this reason, most works have relied on temporal regression [59,60] or pre-established anatomical models [61,62] to learn average temporal trajectories from cross-sectional datasets. We recently presented a method that showed the feasibility of using large cross-sectional datasets to infer personalized pediatric development [23]. However, it assumed specific statistical data distributions and it was not computationally feasible for larger datasets. Our **novel deep phenotyping network will predict personalized development without strong statistical assumptions and could be deployed in state-of-the-art hardware.**

Learning from retrospective outcomes. We have curated a uniquely large retrospective dataset with 1,934 CT images and 1,275 3D photograms that will allow studying abnormal pediatric local head volume development in patients with craniosynostosis. We will leverage our unique dataset to create automated and reproducible methods that will **objectify past clinical experiences and will learn from them** to provide **quantitative insight about pre- and post-surgical head development in patients with craniosynostosis**. This will allow **evaluating and predicting for the first time the short- and long-term treatment outcomes**.

3. Approach

3.1. Team and temporal plan

Table 1. Temporal plan of the project.

SA	Task	Y1		Y2		Y3		Y4		Y5	
1	1										
	2										
	3										
	4										
2	5										
	6										
	7										

Our multi-disciplinary team that has worked together to curate the necessary datasets to ensure the success of this project. Dr. Porras (expert in medical image computing, statistical modeling and machine learning with extensive biomedical training) will coordinate the project, which is a logical continuation of his K99/R00 career development award to study the normal and abnormal changes in the cranial bone shape and thickness in healthy subjects and patients with craniosynostosis. Drs. French (pediatric plastic and reconstructive surgeon and co-director of the Craniofacial Clinic at Children's Hospital Colorado) and Alexander (pediatric neurosurgeon-scientist) will provide the essential clinical knowledge to build realistic models, interpret our results, analyze their implications, and ensure the clinical translation of our tools. Dr. Wrobel (expert in biostatistics and multivariate analysis) will contribute to our multivariate analysis methods, experiment design and statistical analysis. The team will be supported by a postdoctoral researcher with background in medical image computing and machine learning, and we will recruit a computational bioscience graduate student to help with method implementation and software development. The temporal timeline is presented in Table 1. Given our multi-disciplinary expertise, preliminary work, resources, unique data and physical proximity, our team is uniquely positioned to succeed in the execution of this project.

3.2. Preliminary work

3.2.1. Quantification of head shape anomalies from 3D photography

We were the first to develop an automated method to quantify local head malformations in 3D photogrammetry [18,31] for pre- and post-surgical evaluation of craniosynostosis. We used a normative CT image dataset to build a normative statistical model of the head shape using PCA at a single scale, and we quantified local head malformations by comparing the observed head shape of a patient with a personalized normative statistical reference as shown in Fig. 2.

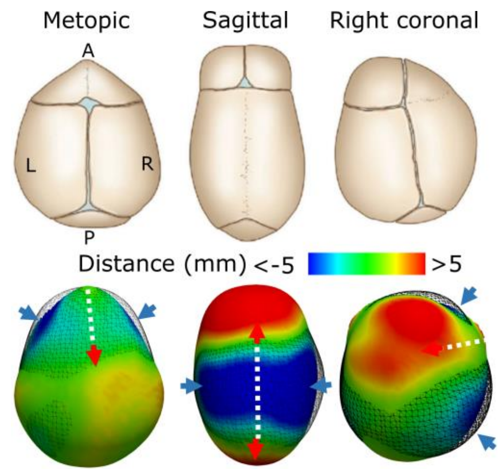

Fig. 2. Standard abnormal phenotype [63] (top row) and local malformations (bottom row) quantified from the 3D photograms of three patients with metopic (left), sagittal (center) and right coronal (right) craniosynostosis. Black wireframes show the personalized normative reference [18] for each patient. Arrows indicate the directions of local volume growth constraints perpendicular to the fused sutures (in blue) and overgrowth parallel to them (in red). Dotted while lines indicate represent fused sutures.

Findings relevant to this proposal: we showed that **CT images and 3D photograms provide the same quantification of shape anomalies** using 467 subjects, and that both can evaluate and identify craniosynostosis with 95% accuracy. We also showed the **potential of 3D photogrammetry to evaluate post-surgical outcomes** [18].

Challenges that will be addressed: Our 3D photogrammetry segmentation methods often needed manual corrections because of the high variability in the data structure and the use of traditional image processing methods. Moreover, they could only evaluate malformations at the scale of a template so they could not quantify volume anomalies. Finally, our models were age- and sex-agnostic so they did not account for the normal anatomical changes during childhood and they could not predict development. In this project, we will create **automatic segmentation methods designed for 3D photogrammetry,** we will build **age- and sex-specific statistical normative references** of local head development, we will **quantify local volume anomalies,** and we will create data-driven methods to **predict personalized temporal anatomical growth.**

3.2.2. Data-driven cranial bone development modeling

We recently created a data-driven cranial bone shape growth model using only cross-sectional CT images of 272 normative subjects (age 0-2 years) [23] . It represented the cranial bone anatomy as a combination of two factors: an age-independent subject representation, and age. Our work was the first fully data-driven model of its kind built without assuming temporal trajectories or growth processes, and its error predicting personalized anatomical changes was as low as the spatial image resolution evaluated on longitudinal normative data.

Findings relevant to this proposal: We showed that **age-independent representations of patient anatomies can be learned from large cross-sectional datasets and used to predict personalized growth**.

Challenges that will be addressed: our methods were not computationally feasible for larger datasets, only considered cranial bones (so they could not be applied to 3D photogrammetry), were sex-agnostic, and used a bilinear PCA formulation that assumed specific data distributions. Hence, we could not predict pathological development. We will design a new method that will learn **age- and sex-independent statistical patient representations without the assumptions and limitation of PCA**, which will be combined with age and sex to **predict anatomical changes both in normative subjects and in patients with craniosynostosis**.

3.3. Data

The three retrospective datasets from Children's Hospital Colorado described below will be used in this project.

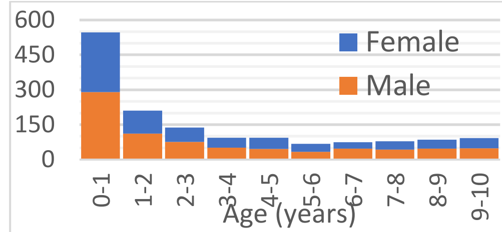

Fig. 3. Number of patients in Dataset A.

Dataset A: cross-sectional normative images

Cross-sectional CT images of 1,480 subjects (687 female, 793 male) without cranial pathology will be used to build a normative statistical model of the head surface. These patients were seen at the emergency department and imaged after trauma. Expert radiological and clinical evaluation discarded pathology or anomalies. Patients with history of cranial pathology, surgery, hardware (e.g., shunts) or genetic syndromes were excluded. The average age is 3.14 ± 3.08 years (range 0-10 years, see distribution in Fig. 3). Higher density of younger patients will allow modeling their faster growth [64]. No inclusion or exclusion criteria related to gender, sex, race, ethnicity or economic resources were used. The racial and ethnic distribution is presented in Table 4.

Dataset B: pre-surgical images of patients with craniosynostosis

721 pre-surgical images (363 CT images and 358 3D photograms) of 468 patients with craniosynostosis (147 female, 321 male, age 0-10 years) will be used to characterize pre-surgical volume anomalies. No selection criteria associated with sex, gender, race, ethnicity or economic resources were used. Higher number of males was expected from incidence reports [65–67]. Patients with previous history of cranial intervention (e.g., shunts) were excluded. Patients with lambdoid craniosynostosis were excluded because of insufficient data of this rare type of synostosis [68–70]. Only patients with fusion of the metopic, coronal and/or sagittal sutures were included. The average age at the last image acquisition before surgery was 7.66 ± 13.70 months. Image and patient distributions per fused suture are presented in Table 2. Racial and ethnic distributions are shown in Table 4.

Table 2. Images / patients per fused suture in Dataset B. Values in brackets show only longitudinal images / patients.

Metopic	132/87 (69/31)
Left unicoronal	32/18 (18/8)
Right unicoronal	46/27 (26/10)
Bilateral coronal	52/22 (42/15)
Sagittal	409/287 (187/86)
Multi-sutural	50 / 27 (29/11)

Longitudinal images: 161 of previous patients (113 male, 48 female) had a total of 371 longitudinal images (154 CT images, 217 3D photograms, 2.30 images per patient) acquired during an average time span of 5.05 ± 11.63 months. Longitudinal images acquired with a time difference lower than one week were not considered.

Dataset C: post-surgical images of patients with craniosynostosis

1,008 post-surgical images (91 CT images, 917 3D photograms) of 302 patients from Dataset B (97 female, 205 male) will be used to quantify post-surgical growth in craniosynostosis. The average age at the first post-surgical image was 1.67 ± 2.00 years and it was acquired an average of 4.94 ± 10.44 months after surgery. The image and patient distributions per fused suture are presented in Table 3, and the racial/ethnic distributions in Table 4. 40 patients underwent endoscopic surgery followed by helmet therapy and had a total of 157 images. 262 patients underwent cranial reconstruction surgery and had a total of 851 images.

Table 3. Images / patients per fused suture in post-surgical Dataset C with endoscopic (Endo) or open reconstruction (Rec.) treatments. Values in brackets show only longitudinal images/patients.

	All		Relapses	
	Endo	**Rec.**	**Endo**	**Rec.**
Metopic	20/4 (20/4)	144/48 (128/33)	0/0 (0/0)	10/2 (10/2)
Left unicoronal	11/4 (10/3)	25/9 (23/7)	8/3 (7/2)	0/0 (0/0)
Right unicoronal	20/4 (20/4)	46/16 (40/10)	7/1 (7/1)	4/1 (4/1)
Bilateral coronal	2/2 (0/0)	43/12 (40/9)	0/0 (0/0)	7/5 (4/2)
Sagittal	86/23 (82/19)	521/157 (487/131)	1/1 (0/0)	48/9 (48/9)
Multi-sutural	18/3 (18/3)	72/20 (99/15)	7/1 (7/1)	20/6 (19/5)

Longitudinal images: 238 patients (159 male, 79 female) had a total of 934 post-surgical longitudinal images (77 CT images, 857 3D photograms, 3.92 images per patient) with an average time of imaging of 1.71 ± 1.26 years. Their first image was acquired 3.58 ± 7.45 months after surgery.

Relapsing patients: 29 patients (10 female, 19 male) with 112 post-surgical images (29 CT images, 83 3D

Table 4. Racial and ethnic distribution of the patients in Datasets A / B / C.

	American Indian / Alaska Native	Asian	Black / African American	Native Hawaiian / Pacific Islander	White	Other / mixed race
Hispanic / Latino	5 / 3 / 1	1 / 2 / 2	4 / 0 / 0	1 / 0 / 0	253 / 69 / 39	218 / 58 / 45
Not Hispanic / Latino	3 / 5 / 2	27 / 3 / 2	186 / 8 / 5	3 / 0 / 0	618 / 284 / 189	161 / 36 / 17

photograms) needed an additional surgery. Their distributions are show in Table 3. Their average age at the first post-surgical image was 2.82 ± 2.75 years and was acquired 3.36 ± 4.56 months after surgery. 23 relapsing patients (13 male, 10 female) had post-surgical longitudinal images, with a total of 106 longitudinal images (26 CT images, 80 3D photograms, 4.08 images per patient) and average time span of imaging of 2.21 ± 1.93 years.

Image resolution: The CT images in our datasets had in-plane resolution of 0.36 ± 0.06 mm with slice thickness of 1.97 ± 1.28 mm. All 3D photograms were acquired using the 3DMD Head Acquisition system (3DMD, Atlanta, GA), which provides higher spatial resolution than our average CT image voxel spacing [71].

3.4. SA 1: To characterize local head volume distributions in craniosynostosis

Rationale: there are no methods that use large datasets to quantify the evolution of the abnormal local volume distributions in patients with craniosynostosis, so the local volume overgrowth mechanisms compensating for their growth constraints have not been quantitatively characterized. Therefore, estimations about local volume development or needs before treatment are based on subjective experience, they are not personalized, and they are often inaccurately described using simple metrics [20,72–74] that do not capture local anatomical variations.

Overview: We will create a statistical anatomical reference of head surface growth using Dataset A. Then, we will create a novel geometric deep neural network specific to 3D photogrammetry to segment the head surface and create standardized representations to overcome the challenges of traditional methods described in Section 2. We will use our normative model to calculate an age- and sex-specific personalized normative reference for each image in Dataset B. We will evaluate local head volume anomalies by comparing the observed anatomies and their normative references similar to Fig. 2 and [18] but accounting for volume, age and sex. Finally, we propose to solve a structural equation model to study the relationships between reduced volume in areas with constrained growth and overdevelopment in other areas, and the role of age and sex in those relationships.

Task 1: Build a model of the anatomical variability of the head surface in the population

Data pre-processing: We will use our existing methods [23] to create a standard anatomical representation (see Fig. 1) of the head surfaces from the CT images in Dataset A. We will combine adaptive thresholding [75] and the marching cubes algorithm [76] to create a mesh representation of the head surface from a CT image. Landmarks at the nasion and tragions (see Fig. 4 (a) for anatomical reference) will be identified by registering all surfaces to a manually annotated reference image. This alignment will also harmonize image pose/orientation. We will use our spherical mapping method [23] to create a standardized 2D anatomical map of the head surface above the plane defined by the nasion and tragions. Hence, the pose-corrected head anatomy of every subject will be represented using a 2D image with uniform size, where each pixel contains the Euclidean coordinates of its corresponding point on the head surface. This will allow for a significant data size reduction representing head surface compared to a CT image, and the use of a pose-standardized representation for all patients [23]. As shown in Fig. 4 (a) and [23], the head anatomy can be accurately reconstructed from these standardized maps.

Model construction: We will adapt previous works [77,78] to build a statistical head surface growth model using the spherical maps of the patients in Dataset A. We will build a PCA model as $X = PC$, where X are the spherical head surface maps after subtraction of the population average, P the principal components and C the coefficients or projections of the spherical maps in the PCA space. As in previous works [77,78], we will select the number of components based on their explained variance. Then, we will regress the average coefficient value for every principal component as $\hat{c}(t, s; \beta_{i,*})$, where t represents age, s represents sex (e.g., 0 for female and 1 for male), and $\beta_{i,*} = \{B_{i,0}, \dots, B_{i,M}\}$ are the M regression parameters for principal component i. We will also regress the standard deviations $\hat{\sigma}(t, s;\ A_{i,*})$, where $A_{i,*} = \{A_{i,0}, \dots, A_{i,L}\}$ are the L parameters modeling the standard deviation for principal component i. The regression parameters of the age- and sex-specific average and standard deviation that represent normal distribution of the population in each principal component i will be estimated as:

$$\beta_{i,*} = \arg\min\left(\frac{1}{N}\sum_{j=1}^{N} \omega(t_j)\left(c_{ij} - \hat{c}(t_j, s_j; \beta_{i,*})\right)^2\right), \text{ and}$$

$$A_{i,*} = \arg\min\left(\frac{1}{N}\sum_{j}^{N}\omega(t_j)\left(\sqrt{\left(c_{ij} - \hat{c}(t_j, s_j; \beta_{i,*})\right)^2} - \hat{\sigma}(t_j, s_j;\ A_{i,*})\right)^2\right),$$

where N is the number of subjects in Dataset A, and c_{ij} is the coefficient representing subject j in principal component i. $\omega(t_j)$ is a weighting term that will compensate for variable data density at different ages as $\omega(t_j) = \left(\frac{1}{N}\sum_{k=1}^{N} K(t_j - t_k)\right)^{-1}$, where K is a Gaussian kernel. We will evaluate different regression functions such as polynomials or exponential functions, and $A_{i,*}$ and $\beta_{i,*}$ will be estimated by regular gradient descent optimization.

Task 2: Create a geometric neural network to automatically segment the head surface from 3D photogrammetry

Data pre-processing: 982 3D photograms from Datasets B and C were manually annotated by an expert radiographer with the nasion and tragions, which we used to extract the head surface in Dataset A. We will use the location of those landmarks in the annotated photograms to create a standard spherical map representing their head surface as we did in Task 1, which we will use to train automatic 3D photogram segmentation methods.

Head surface segmentation from 3D photogrammetry: We propose a new neural network that will automatically create standard spherical map representations of the head surface given a 3D photogram. Traditional image-based spatial convolutions are not applicable to photogrammetry because, unlike typical images, it has variable numbers of connected vertices with no correspondences. Since a 3D photogram can be regarded as a graph in which vertices represent nodes and their triangulations edges [79], we propose to learn features $\{h_i\}$ at each vertex by calculating spatial convolutions in the spectral domain of the graph using Chebyshev polynomials [79,80]. Given the location $\boldsymbol{x}$, the RGB color $\boldsymbol{a}$ and surface normal unitary vector $\boldsymbol{n}$ at vertex i in the 3D photogram of a patient, a convolution operation using Chebyshev polynomials can be calculated as:

$$h_i = \sum_{k=0}^{K-1} \theta_k T_k(\tilde{L}) b_i\ ,$$

where $\tilde{L}$ is the normalized graph Laplacian calculated from the graph adjacency (A) and degree (D) matrices as $\tilde{L} = D^{-\frac{1}{2}}AD^{-\frac{1}{2}}$ [80], $T_k(\tilde{L})$ are the k^{th} order Chebyshev polynomials, b_i the i^{th} input vertex feature ($\boldsymbol{x}$, $\boldsymbol{a}$ and $\boldsymbol{n}$ in the first layer) and θ_k the trainable weights used to calculate the features h_i.

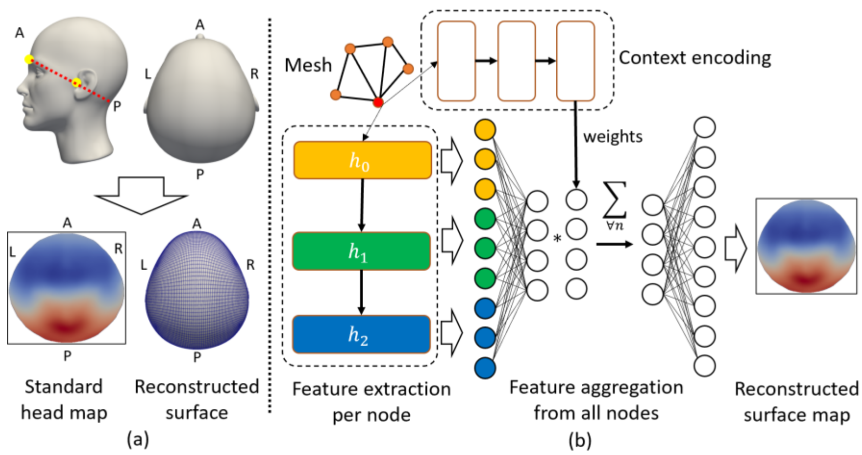

Fig. 4. (a) Top: nasion and tragion as yellow circles defining the plane used to segment the head surface (shown as a doted red line). Bottom: standard anatomical head surface map where pixels contain the Euclidean coordinates of the points on the head surface. (b) Neural network to create standard anatomical head maps from 3D photogrammetry. Squares represent spectral convolutional layers and circles fully connected layers. h_i represent multi-resolution convolutional features.

We propose the novel geometric autoencoder-like [81] network shown in Fig. 4 (b) to quantify multi-resolution features using spectral convolutions. A new aggregation scheme will add features from every vertex using normalized weights that will be learned from the context around them, which will allow masking out vertices that are not relevant to represent the head surface (e.g., lower face or neck). The result will be a latent representation of the head anatomy of a subject that is independent from the number of vertices in the photogram. Dense layers will use this latent vector to create a standardized head surface map. All layers will use ReLU activation except for the output layer.

Training: We will split our dataset randomly in 80%, 10% and 10% for training, validation and test, respectively. We will repeat this process until the method has been evaluated using all data. We will augment [82] our dataset using random translations, rotations, scalings, and resampling operations using quadric error decimation [83] to reduce training overfitting. Note that although our spherical map representation is invariant to rotation or translation, the input 3D photograms are not. To estimate the network parameters, we will minimize the mean squared difference between the reconstructed head surface maps and the ones obtained in pre-processing stages from manual annotations. We will evaluate different optimizers (e.g., stochastic gradient descent, Adam, etc.), tune learning rates and momenta empirically, evaluate regularization approaches (e.g., L1 or L2) and use batch normalization and dropout to improve model performance and generalizability.

Task 3: Characterize patterns of abnormal local volume distributions in craniosynostosis

Quantification of local volume anomalies: We will use previous neural network to create anatomical head surface maps for all images in Dataset B. Since normative PCA models provide an accurate reference to quantify local anomalies [13,14,18,32], we will calculate the coefficients representing these images in the PCA space of our normative model (Task 1). Then, as in previous work [18] and shown in Fig. 2 with a volume-, age- and sex-agnostic model, we will assume a normal statistical distribution in the healthy population to calculate for every patient a personalized normative anatomical reference. Specifically, we will constrain their PCA coefficients as $c_{ij} \in \left[\hat{c}\left(t_j, s_j; \beta_{i,*}\right) \pm 3\hat{\sigma}\left(t_j, s_j;\ A_{i,*}\right)\right]$, where c_{ij} is the PCA coefficient of patient j in principal component i, t_j its age and s_j its sex. We will use the constrained coefficients to reconstruct a personalized normative anatomical reference and we will calculate the local signed distances between the patient anatomies and their normative references in *mm*, as color-coded in Fig. 2. Note that unlike previous shape models built at a single scale, the spatial integration of these distances across the head surface will now provide a quantification of the local volume anomalies in mm^3, where the sign indicates lack (negative) of excess (positive) of local volume.

Modeling common abnormal volume patterns: We aim to quantify the relationships between local head volume loss in areas with growth constrained by suture fusion and compensatory overdevelopment in other areas using Dataset B. These patterns are common between patients with fusion of the same suture [85–87] as shown in Fig. 2, which enabled our previous methods to identify which suture was fused in 99% of patients [18].

We propose to model the common patterns of volume anomalies as the result of the contribution of latent factors that represent the phenotype associated with the fusion of each suture. We will approach this problem using confirmatory factor analysis to identify how each latent factor contributes to the specific volume anomalies quantified in Dataset B. Unlike PCA, factor analysis will allow identifying commonly observed correlations between the areas of volume loss and gain (communalities) separately from the anatomical variability specific to every patient (uniqueness). We propose a structural equation model of the local volume anomalies in Dataset B as $\boldsymbol{Y} = \boldsymbol{L\Lambda} + \boldsymbol{E},$ where Y are the local quantified volume anomalies on the head surface maps, E the patient uniqueness, $L\Lambda$ represents the communalities, L the latent factors and Λ the loading weights. Fig. 5 presents the structural equation model diagram. To avoid normality assumptions of maximum likelihood methods fitting a model of pathology, we will fit our model to the data using the robust diagonally weighted least squares method [88,89]. We will also evaluate factor rotation [90,91] to simplify the model structure and increase its explanatory ability.

Data aggregation and model identification: We will integrate the local distances between the head surfaces in Dataset B and their normative references at the 12 regions shown in Fig. 1 to quantify volume anomalies in mm^3. As in previous work [18] and benefitting from the anatomical consistency in the population, we will calculate the average bone and suture locations using the segmentations from the CT images of Dataset A obtained from our existing methods [23]. We will project this average onto the head maps from Dataset B to approximate the location of their bones and sutures, since they are not visible in 3D photogrammetry. Then, we will partition the bone regions based on the distance to the closest suture as shown in Fig. 1. This will allow differentiating local volume anomalies at 12 anatomical regions, so our model will consider 12 variables per patient in matrix $\boldsymbol{Y}$. Since Dataset B contains patients with metopic, sagittal, left and/or right coronal craniosynostosis, our model will have four latent factors ($\boldsymbol{L}$).

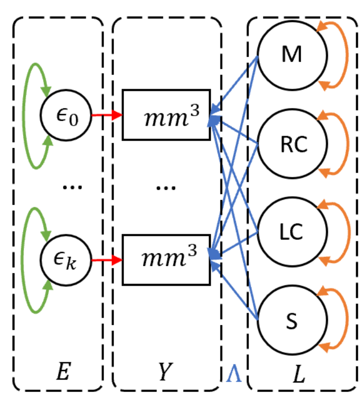

Fig. 5. Structural equation model diagram [84] where volume anomalies are explained by latent factors associated to the early fusion of the metopic (M), right or left coronal (RC, LC), or sagittal sutures (S).

The model implied covariance matrix is $\boldsymbol{\Sigma}(\boldsymbol{\theta}) = \boldsymbol{\Lambda\Psi\Lambda}^{\mathbf{T}} + \boldsymbol{\Theta}_{\boldsymbol{\epsilon}}$, where $\boldsymbol{\theta}$ are the model parameters, $\boldsymbol{\Psi}$ the latent factors ($\boldsymbol{L}$) covariance matrix, and $\boldsymbol{\Theta}_{\boldsymbol{\epsilon}}$ the covariance matrix of the residuals or uniqueness ($\boldsymbol{E}$). Since the covariance matrices are symmetric, the number of model parameters is 136 (48 from loadings, 10 from latent factors, and 78 from residuals). Since the residuals are independent, we will assume their covariances to be zero (66 parameters). Hence, there are 70 ($136 - 66$) free parameters. No assumptions on factor independence will be made. Since 78 parameters (from 12 variables) are known from the sample covariance matrix, our model is over-identified with 8 ($78 - 70$) degrees of freedom [92,93]. After parameter estimation [88,89], we will have a quantitative model of the relationships between the volume anomalies at different areas explained by the fusion of individual sutures.

Severity of volume anomalies associated with age and sex: We will calculate the factor values that explain the anomalies in Dataset B as $\boldsymbol{L} = \arg\min \|\boldsymbol{Y} - \boldsymbol{L\Lambda}\|^2$. Then, similar to Task 1, we will create a regression model of the average and standard deviation of these factor values as function of age and sex grouped by single fused sutures. These regressions will quantify the variability of the phenotypical severity in the population with age and sex for each fused suture. We will quantify statistical differences in the severity of different age and sex groups, and we will use path analysis in our structural equation model [94–96] to explore covariances and correlations between volume anomalies at different head regions.

Task 4: Integration of methods on prototype for clinical translation

We have built a graphic user interface called CU-MIA (Colorado University - Medical Image Analysis) for fast prototyping and clinical translation of our methods. It currently integrates our cranial and head shape analysis for use at the Craniofacial Clinic of Children's Hospital Colorado. Our new methods to segment and analyze 3D photogrammetry, to quantify and predict local head volume anomalies and to evaluate the severity of craniosynostosis phenotypes will be integrated in CU-MIA using Python with simpleITK, VTK and OpenGL. We will also extend our interface with the methods from the second specific aim, which will enable pre-and post-surgical patient evaluation at the clinic.

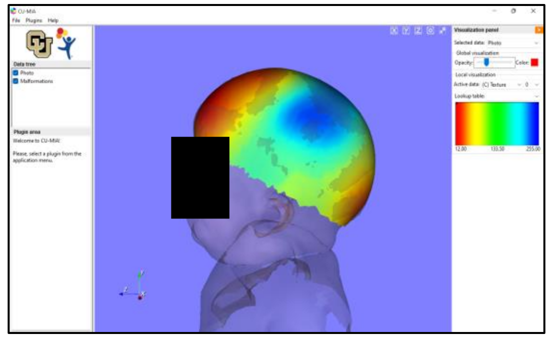

Fig. 6. Interactive head shape analysis from a 3D photogram in CU-MIA.

Evaluation and benchmarks of success for specific aim 1

• Normative head surface statistical model (Task 1): We will use validated state-of-the-art methods [77,78] to construct our normative model. We will evaluate success of our regressions using F-tests and a p-value threshold of 0.05. We will also evaluate goodness-of-fit metrics based on the normalized root mean squared error, adjusted R^2, and the predicted R^2 (calculated using 20-fold cross-validation).

• Head surface segmentation (Task 2): We will use 10-fold cross-validation on the 982 photograms with manual annotations. We will evaluate our methods based on the Euclidean distances between the calculated head surface maps by our network and the ones obtained from manual annotations. We will consider success if errors are equal or lower than our CT image resolution (2.03 mm) evaluated using Mann-Whitney U-test (for p=0.05).

• Identification of volume distribution patterns (Task 3): We will evaluate our model using a χ^2 test with p-value threshold of 0.05. Given its the tendency to reject good models built from large datasets [97], we will also calculate the root mean squared error of approximation [98] (we will aim for values lower than 0.06 [99,100]) and the goodness of fit index [100] (we will aim for values higher than 0.9 [101]). The regression models of age and sex will be evaluated using F-tests with a p-value threshold of 0.05, and their predictive accuracy will be assessed using the longitudinal images in Dataset B. We will consider success if the latent factor values calculated for different longitudinal images of the same patient and normalized to the age- and sex-specific regressed average and standard deviation are not significantly different (for p=0.05 using a Wilcoxon test), which means that our model captures temporal changes. Age and sex differences will be evaluated using a Mann-Whitney U-test (for p=0.05).

Challenges and alternative strategies for specific aim 1

• Normative head surface statistical model (Task 1): we don't expect methodological challenges using previous methods [77,78]. We have used 272 subjects to successfully create a cranial bone shape model between birth and two years [23] that achieved predictive accuracies within the CT image resolution ranges. At least 60% of head growth in the first 10 years of life occurs during the first two years [21]. Hence, based on our previous study [23], we would need 181 patients (272 x 0.4 / 0.6) to model growth patterns in the age range 2-10 years. Given that our dataset has 758 patients in the range 0-2 years and 580 in the range 2-10 years, we do not foresee sample size-related issues. Although we would have liked incorporating race in our model, our homogeneous local population will not allow it (see Table 4). As in our previous works [102–109], we are committed to continue seeking collaborations and working towards increasing the diversity of our datasets to reduce racial and ethnic biases.

• Head surface segmentation (Task 2): We do not expect problems using Chebyshev polynomials, which have been shown to accurately quantify spatial convolutions [79,80]. We also proposed significant data augmentation and approaches to control overfitting during optimization. Since we have shown that CT images and 3D photogrammetry provide similar anatomical quantifications [18], we expect success in this task.

• Quantification of volume distribution patterns (Task 3): Patients with early fusion of the same suture present the same phenotype with diverse severity [85–87]. For that reason, our structural equation model represents phenotypes explained by factors with variable magnitude associated to each cranial suture. We intentionally avoided assumptions about factor covariates in our model because these relationships have not been explored. Our model is overidentified and its construction is feasible. Although there is no consensus on approaches for sample size estimation, the most common reference is a minimum ratio of 10:1 between observations and factors

[90]. Given our dataset size and number factors, we are not concerned about sample size. Although unexpected from existing evidence [85–87], we may find large discrepancies between the latent factor magnitudes explaining the phenotype of patients with the same age, sex and fused suture. In that unlikely scenario, we may consider other approaches to model fitting (e.g., maximum likelihood) and/or adding constraints to model fitting. Specifically, we may consider regularization terms that promote minimal contributions from factors associated with sutures that are not fused in specific patients. This will ensure that their anomalies are only explained by factors associated with their fused sutures. If our model fit metrics do not meet our success criteria, it will mean that there are other important factors that need to be considered to explain local volume anomalies. Hence, we could consider using exploratory factor analysis techniques [110–113] to uncover such factors.

Outcomes of specific aim 1: upon completion of this specific aim, we will have created geometric learning methods to create standard anatomical representations from 3D photogrammetry. We will have built the first age- and sex-specific model of head surface development and methods to quantify local volume anomalies. Moreover, we will have created a structural equation model that explains and quantifies volume anomalies in craniosynostosis as the contribution of latent factors associated with the fusion of individual sutures. This model will provide the first explicit quantification of the correlations between local volume underdevelopment caused by growth constraints and the compensatory volume overdevelopment mechanisms. Finally, we will have built a regression model of phenotype severity as a function of age and sex. This will enable studying and predicting the progression of local volume anomalies, which could have important implications in treatment planning.

3.5. SA 2: To predict and evaluate local head volume growth after treatment of craniosynostosis

Rationale: because of the lack of quantitative methods to evaluate surgical results in patients with craniosynostosis, treatments and outcomes are highly diverse [3–6]. Although we showed that 3D photogrammetry can quantify post-surgical head malformations [18], our methods did not consider age, sex or volume, so they could not evaluate or predict growth. We will leverage our datasets to build methods to quantitatively evaluate, compare and predict post-surgical development in patients with craniosynostosis who underwent different treatments.

Overview: We will create a deep learning method trained with the images from Datasets A and B to learn age- and sex-independent latent representations of patient anatomies. These will be combined with age and sex to make personalized anatomical predictions. We will predict growth from the first post-surgical image for the patients in Dataset C with longitudinal studies. We will quantify abnormal growth as the difference between predictions and observations, and we will quantify volume anomalies at every longitudinal image. We will regress and compare these two metrics between patients who underwent endoscopic release and cranial reconstruction surgery, and we will study differences associated with age and sex. Finally, we will train a neural network using Dataset C to evaluate and predict post-surgical growth, and we will assess its potential to detect of relapses.

Task 5: Create a deep phenotyping model predictive of head development

We propose the new neural network architecture presented in Fig. 7 with three components: a generative adversarial network (GAN) base architecture [54,55], an autoencoder [81] and Siamese layers [114–116]. Its goals are two-fold: make growth predictions for specific patients ("predicted" map in Fig. 7) and learn population distributions to generate synthetic anatomies ("synthetic" in Fig. 7). The network inputs will be the head maps from Datasets A and B, age, sex (binarized), and a binary vector indicating the fusion of each cranial suture.

Generative adversarial architecture: following the GAN design that has shown great performance learning data distributions to generate synthetic data [51,55], we propose a generator that will reconstruct realistic head surfaces from randomly generated Gaussian noise, patient age, sex and information about suture fusion. These data will be provided to fully connected layers whose output will be reshaped into 2D images and used by up-sampling convolutional layers to reconstruct synthetic head maps. Note that noise generation is independent from patient demographics or pathology. A discriminator will compete with the generator to distinguish real from synthetic maps as in [54], and will also evaluate patient age, sex and fused suture for classification. Training will be done by solving the following minimax problem [54]: $\min_G \max_D V(D,G) = E_{\boldsymbol{x}\sim p_{data}(\boldsymbol{x})}[\log D(\boldsymbol{x},\boldsymbol{y})] + E_{\boldsymbol{z}\sim p_{\boldsymbol{z}}(\boldsymbol{z})}\left[\log\left(1 - D(G(\boldsymbol{z},\boldsymbol{y}),\boldsymbol{y})\right)\right]$, where $\boldsymbol{x}$ is a head surface map, $\boldsymbol{z}$ represents Gaussian noise with probability distribution $p_{\boldsymbol{z}}(\boldsymbol{z})$, $\boldsymbol{y}$ encodes age, sex and suture fusion information, $G(\boldsymbol{z},\boldsymbol{y})$ is a synthetic head surface map created by the generator G, and $D(\boldsymbol{x},\boldsymbol{y})$ is the probabilistic prediction by the discriminator D evaluating image $\boldsymbol{x}$ and patient data $\boldsymbol{y}$. Using this formulation, the parameters of the discriminator will be trained to maximize the probability of correctly classifying synthetic and real maps, while the parameters of the generator will be trained to minimize that probability.

Autoencoder: An autoencoder will learn latent representations of the head anatomy of a subject to reconstruct its standard head surface map (orange box in Fig. 7). Note that, unlike traditional autoencoders, age, sex and suture fusion data will be provided to the up-sampling layers (in red) together with the latent representation of the patient. As explained below, this will enable learning age-, sex- and pathology-agnostic representations. The auto-encoder will be trained by minimizing the mean squared distance between the predicted and observed head maps as $A(P) = \min_{P} \|P(\boldsymbol{x},\boldsymbol{y}) - \boldsymbol{x}\|^2$, where $P(\boldsymbol{x},\boldsymbol{y})$ is the predicted map from the image $\boldsymbol{x}$ and data $\boldsymbol{y}$ (age, sex, fused suture).

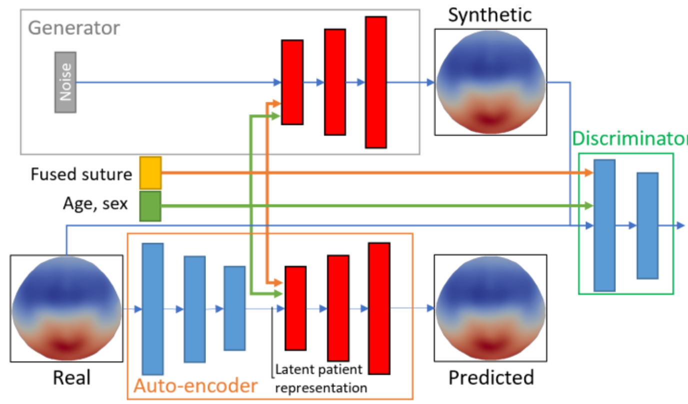

Fig. 7. Proposed deep phenotyping architecture to predict patient-specific anatomical changes.

Siamese layers: Learning how age affects the anatomy of specific patients is challenging without longitudinal observations. Although age, sex and pathology are provided to the up-sampling layers of our autoencoder to make anatomical reconstructions, these layers will not necessarily use that information to reconstruct specific patient anatomies, since these data could also be inferred from the real images by the down-sampling layers. Hence, a traditional autoencoder cannot learn how age modifies specific patient representations only from cross-sectional data. Separately, our generator is trained to learn how age, sex and pathology modify random normally distributed values (the noise) to create anatomies that resemble real observations. To benefit from that behavior, we propose the up-sampling layers of the autoencoder (in red) to be Siamese [114–116] to the ones in the generator and they will share the same parameters. This will have two effects: (1) it will make the down-sampling layers of the autoencoder create similar data distributions to the Gaussian noise in the generator, which is independent from age, sex and pathology; and (2) it will make the up-sampling layers of the autoencoder only use the age, sex and pathology data provided to it, just like the generator does. Hence, the autoencoder will learn how to create age-, sex- and pathology-agnostic patient representations that can be combined with age, sex and pathology data to predict specific anatomies. Given one observation of a patient's anatomy from one image, our autoencoder will then be able to predict its personalized anatomy at any other age that is provided to its up-sampling layers. All layers will use ReLU activation except for the output layers. Softmax activation will be used in the discriminator output.

Training: Our neural network will be trained to reconstruct all images in Datasets A and B. Following typical GAN-based training [54], our model will be trained iteratively using a gradient descent-based optimizer, taking two steps at every iteration. First, the parameters of the generator and auto-encoder will be jointly updated to minimize $V(D,G)$ and $A(P)$. Second, the parameters of the discriminator will be updated to maximize $V(D,G)$. The weights balancing the contribution of $V(D,G)$ and $M(P)$ will be chosen empirically. We will evaluate regularization (e.g., L1 or L2), batch normalization and dropout to improve performance and generalizability.

Task 6: Quantify and compare post-surgical anomalies after different surgical treatments of craniosynostosis

We aim to answer two questions: (1) how do volume anomalies improve after different surgeries? and (2) how do patients develop after different treatments? Because of the many factors affecting post-surgical growth, it would be challenging to gather enough longitudinal data to build post-surgical growth models for every combination of those factors at every anatomical location. Instead, we will quantify global metrics of the overall evolution of volume anomalies and growth after surgery using Dataset C as described below.

• Change of volume anomalies: We will use our normative statistical head surface model (Task 1) to calculate local volume anomalies for all images in Dataset C, which we will integrate spatially in 12 anatomical regions as in Task 3. We will normalize the local volume $v_{r,k}$ calculated from image *k* and region *r* by age and sex as $\bar{v}_{r,k} = v_{r,k}/\hat{v}(t_k, s_k)$, where $\hat{v}(t_k, s_k)$ is the regressed standard deviation of the local volume anomalies in Dataset A at region r for subjects with age t_k and sex s_k. We will calculate a global anomaly index $a_k = \sum_{\forall r} |\bar{v}_{r,k}|$ from each image *k* that will measure the overall normalized severity of volume anomalies. We will group all images by type of treatment (157 endoscopic vs. 851 cranial reconstruction) and we will linearly regress our age- and sex-normalized index as a function of time from surgery t_k. For each surgical group g, we will calculate regression parameters a_g(intercept) and b_g(slope) as $a_g, b_g = \arg\min \sum_{\forall k} w_k \left(a_k - (a_g + b_g t_k)\right)^2$, where w_k is a weighting term that will be used to balance the contribution of images from patients with different number of studies.

• Abnormal post-surgical growth: We will use the model form Task 5 to predict normal growth from the first image of the patients who have longitudinal studies in Dataset C to the age of every subsequent image (setting all suture fusion vector elements to zero). This will establish a personalized normal growth reference for every patient after their first post-surgical image. Then, we will calculate the local Euclidean distance between the longitudinal observations and the personalized predictions in mm. We will integrate these differences spatially in the 12 anatomical regions shown in Fig. 1 to quantify local abnormal volume growth in mm^3. All values will be divided by the normative expected volume from the predictions made by our neural network to obtain normalized values. Finally, we will create an abnormal growth index d_j for every longitudinal observation k as $d_k = \sum_{\forall r} |\bar{d}_{r,k}|$, where $\bar{d}_{r,k}$ is the normalized volume growth at region r. Similar to our anomaly indices a_k, we will group our patients per surgical treatment and we will regress d_k as a function of time after surgery.

These indices will enable comparing surgical approaches in terms of global improvement of volume anomalies and abnormal growth. We will use t-statistics to evaluate significant differences between surgical groups.

Task 7: Predict post-surgical development and evaluate treatment outcomes

Post-surgical development prediction: We will base on an auto-encoder [81] architecture (orange box in Fig. 8) to build a convolutional neural network that predicts anatomical changes given the post-surgical image of a patient, sex, ages at image acquisition and surgery, a binarized vector indicating what suture was fused before surgery, a binary value indicating the type of surgery (endoscopic vs. reconstruction), and the growth time in days. We will train the network using the longitudinal post-surgical images in Dataset C (238 patients with 3.92 longitudinal images/patient). For each patient, we will generate all possible combinations of post-surgical image pairs and build a dataset with 1,590 post-surgical longitudinal image pairs. During training, the first image from each pair will be used as input and the second will be used as ground truth for the temporal anatomical predictions.

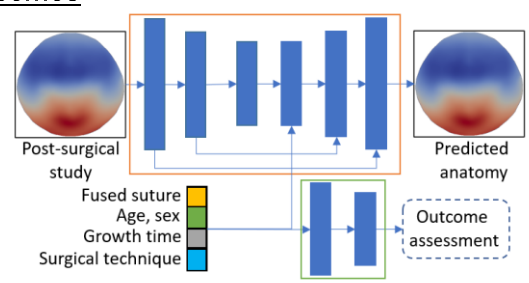

Fig. 8. Neural network to predict post-surgical growth and evaluate outcomes.

Evaluation of treatment outcomes: We will also train our neural network to evaluate treatment outcomes. To benefit from the higher accuracies shown in joint-task learning [117], we will embed an outcome evaluation module (green box in Fig. 8) into our post-surgical growth prediction network to achieve two tasks: predict the volume anomaly index from Task 6 to quantify evolution of abnormal volume distributions, and estimate the risk of relapse of the patient. We will use the previously calculated index a_k at the time of the second image in the pair, and the patient history data about additional surgical treatments from Dataset C as ground truth to train these two additional tasks. Except for the output layer, all layers will be followed by ReLU activation.

Training: the network will be trained using pairs of images from Dataset C as previously explained. Different gradient descent-based optimizers will be evaluated to minimize a loss function with three weighted terms: the mean squared difference between the predicted anatomical map and the second image in every pair, the mean squared difference between the predicted anomaly index and the one calculated from the second image in every pair, and a binary cross-entropy loss between the softmax-activated relapse risk estimation and the known labels (1 for relapsing patients and 0 otherwise). We will use sample weighting to compensate for lower number of relapsing patients. We will tune the number of layers and parameters, use batch normalization and dropout as necessary to optimize performance. Relative weights between loss terms will be tested empirically.

Evaluation and benchmarks of success for specific aim 2

- Predictive deep-phenotyping model (Task 5): We will use 10-fold cross-validation to evaluate our network using Datasets A (N=1,480) and B (N=721), preserving the balance between population groups (age, sex), pathology and number of images per patient between folds. Single image reconstructive accuracy will be evaluated using mean squared Euclidean distances between inputs and reconstructions. The predictive accuracy will be evaluated using the longitudinal images in Dataset B. For each test patient with longitudinal images, we will generate all combinations of image pairs and predict growth between the time of the two images (using the first acquired image as input and setting the predictive age to the patient's age at the time of acquisition of the second image). All comparisons will be quantified as the mean local Euclidean distance between the observations and predicted anatomical maps. We will consider success if our method provides accuracies equal or lower than the CT image dataset resolution (2.03 mm) evaluated using Mann-Whitney U-test and p-value threshold of 0.05.

- Quantification of post-surgical growth between treatments (Task 6): We will compare the regressions between surgical groups using t-tests [118] and a p-value threshold of 0.05. Given the heterogeneous findings from studies comparing endoscopic release and cranial reconstruction surgeries based on insufficient patient-specific anatomical growth data [119–125], we will consider any results an important scientific contribution.

• <u>Post-surgical growth prediction and outcomes evaluation (Task 7)</u>**:** We will use 10-fold cross-validation to evaluate predictions using the longitudinal images in Dataset C. For each patient, we will generate combinations of post-surgical image pairs and build a dataset with 1,590 pairs. During training, the first image will be used as input and the second as ground truth for the predictions. Differences between the predicted anatomies and observations will be evaluated in terms of mean Euclidean distances. We will consider success if we achieve errors equal or lower than the CT image dataset resolution (2.03 mm) evaluated using Mann-Whitney U-test with a p-value threshold of 0.05. Predictions in the anomaly indices will be considered successful if the average prediction error is lower than the standard deviation in the population. Finally, we will evaluate the feasibility identifying relapsing patients in terms of accuracy, sensitivity and sensitivity. Since there are no previous studies to identify relapses early from observed post-surgical anatomies, we consider this classification as exploratory.

Challenges and alternative strategies for specific aim 2

• <u>Predictive deep-phenotyping model</u> (Task 5): Each module has been extensively used in other works, so we do not expect implementation challenges. We have shown in our previous work [23] that cross-sectional data can be used to make personalized growth predictions. If limited temporal consistency is observed in our predictions, we will consider adding longitudinal image pairs-based training (benefitting from the longitudinal images in Dataset B, similar to Task 7) to our single-image training approach. Although there is no agreement on pre-hoc methods to determine sample size in convolutional networks [126], we are confident about our sample size. Others have trained GAN-based architectures with fewer and more complex images [45]. Additionally, Siamese modules have shown great performance with limited size and imbalanced datasets [114]. If more data are needed, we will explore traditional and deep learning-based generative data augmentation approaches [127–129].

• <u>Quantification of post-surgical growth between treatments</u> (Task 6): We do not foresee significant issues with Task 6. Given the lack of studies of post-surgical anatomical development in the literature [119–125], we will consider our results comparing both surgical approaches as an important contribution to the scientific community. We may find higher variability of our global metrics in the population who underwent open cranial reconstruction compared to endoscopic treatment because of the diversity of surgical sub-techniques (i.e., different approaches to perform osteotomies and reconstructions). Variable findings will prompt a study comparing sub-techniques.

• <u>Post-surgical growth prediction and outcomes evaluation (Task 7)</u>**:** Given the demonstrated performance of autoencoder-based networks to predict changes [130], the implementation should not be a problem. We proposed a relatively simple architecture to minimize overfitting. If we do not meet our success criteria, we will explore more complex architectures with similar goals such as transformers [131,132]. We will also benefit from this architecture to train our outcome evaluation module through multi-task learning [117]. We may, however, not obtain high accuracies predicting relapses, since the decision to receive additional treatments is highly subjective (e.g., patients with similar outcomes may choose differently to receive additional treatment). For that reason, we consider our prediction of relapses as exploratory. High accuracy will prompt seeking collaborations to increase our data pool and investigate potential reasons of relapse. Low accuracies may mean that the subjectivity of the decision to receive additional treatments may not be highly correlated with post-surgical anatomical growth.

Outcomes of specific aim 2: We will have developed a new method to create age- and sex- independent patient representations and predict development both in normative subjects and patients with craniosynostosis. We will have used it to quantify and compare for the first the post-surgical growth and the evolution of volume anomalies in patients who received endoscopic and cranial reconstructions surgical treatments. Finally, we will have created a new neural network that can evaluate and predict post-surgical development and outcomes after treatment of craniosynostosis, which is essential to objectify the evaluation of different surgical techniques.

4. Conclusion

This proposal is a continuation of Dr. Porras' K99/R00 Pathway to Independence project, and it addresses specific gaps learned from that experience. This project will further our understanding of the correlations between local cranial volume growth restrictions and compensatory overgrowth mechanisms in craniosynostosis, which is important to quantify and predict local volume needs for treatment planning. We will also study differences in those relationships associated with age and sex, and we will quantify for the first time how different surgical treatments affect growth and the progression of volume anomalies. Our methods to evaluate, compare and predict surgical outcomes will be embedded in a software interface for clinical translation and our findings exploring the identification of relapsing patient could potentially reduce the morbidity associated with suboptimal treatments. Moreover, our innovative predictive deep learning approach could have applications to quantitatively study other pediatric developmental pathologies. Finally, our methods will advance the clinical utility of non-invasive 3D photogrammetry, which could help reduce the use of harmful radiation and sedation in children.

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